Logistic regression on k classes

best Method → Softmax

lets see how Softmax Method works

Softmax

K classes

Input → $X \in R^{m \times n}$, $Y \in R^{m \times 1}$

unique classes = np.unique(Y)

k = len(unique classes)

eg: $Y = \begin{bmatrix} 6 \\ 2 \\ 1 \\ 36 \\ 6 \\ 1 \end{bmatrix}$

$\text{uni} = [6\ 2\ 1\ 36]$

$k = 4$ $\downarrow k_1 \downarrow k_2 \downarrow k_3 \downarrow k_4$

lets do one hot encoding on Y such that they can be presented as a linear score for different classes

$$Y = \begin{bmatrix} 6 \\ 2 \\ 1 \\ 36 \\ 6 \\ 1 \end{bmatrix} \rightsquigarrow Y = \begin{array}{cccc} k_1 & k_2 & k_3 & k_4 \\ \hline 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array}$$

now probabilites for individual and class be represent as bernoulli score

$$\underset{R^{m \times 1}}{Y} \longrightarrow \underset{R^{m \times k}}{Y\_k}$$

$Y\_k(i,j) \in \{0, 1\}$

we have m samples in total

Assume: any sample with a particular k class can be represented as a linear score w.r.t particular class

$Y \in R^{m \times k}$

$X \rightarrow$ add 1s column of intercept

$$X = \begin{bmatrix} 1 & X_1^{(1)} & X_2^{(1)} & X_3^{(1)} & \cdots & X_n^{(1)} \\ 1 & X_1^{(2)} & X_2^{(2)} & X_3^{(2)} & \cdots & X_n^{(2)} \\ \vdots & \vdots & & & & \vdots \\ 1 & X_1^{(m)} & X_2^{(m)} & \cdots & & X_n^{(m)} \end{bmatrix} \in R^{m \times d}$$

$d = n+1$

for a particular sample $X^{(i)} = [1 \; X_1^{(i)} \; X_2^{(i)} \cdots X_n^{(i)}] \in R^{1 \times d}$

$Y \in R^{m \times k}$ $\quad Y_{ij} = \{0, 1\}$

softmax differes from one-vs-rest in the way of its way of optimize the parameter of β for each class fit (linear) at same time such that they are dependent on other rather then individual

class 1 → $\beta_1 = [\beta_0^{(1)} \; \beta_1^{(1)} \; \beta_2^{(1)} \cdots \beta_n^{(1)}]^T \in R^{d \times 1}$

class 2 → β_2

class k → $\beta_k = [\beta_0^{(k)} \; \beta_2^{(k)} \cdots \beta_n^{(k)}]^T$

lets make a whole β_k that represents all β for each clas

$$\beta_k = [\beta_1 \; \beta_2 \; \beta_3 \cdots \beta_k]$$

$$\beta_k = \begin{bmatrix} | & | & & | \\ \beta_1 & \beta_2 & \cdots & \beta_k \\ | & | & & | \end{bmatrix} \in R^{d \times k}$$

$$Z = X\beta_k \in R^{m \times d} \times R^{d \times k} = R^{m \times k}$$

$$Z_{ij} = X^{(i)} \beta_j$$

assumption of score to be linear

$x^{(i)} = [1 \; x_1^{(i)} \cdots x_n^{(i)}] \quad R^{1 \times d}$

$\beta_j = [\beta_0^{(j)} \; \beta_1^{(j)} \cdots \beta_n^{(j)}]^T \quad R^{d+1}$

$X^{(i)} \beta_j \in R^{1 \times 1}$

in code

$(X^{(i)}\beta_j)$.item()

or

float$(X^{(i)}\beta_j)$

we know the softmax fun for each

$$P_{ij} = \frac{\exp(Z_{ij})}{\sum_{l=1}^{K} \exp(Z_{il})} = \frac{e^{x^{(i)}\beta_j}}{e^{x^{(i)}\beta_1} + e^{x^{(i)}\beta_2} + \cdots + e^{x^{(i)}\beta_k}}$$

probability of ith sample belonging to the jth class

$$P_i = [P_{i1} \; P_{i2} \; P_{i3} \cdots P_{ik}]$$

$$\boxed{\underset{k}{\operatorname{argmax}} \, P_i}$$

prediction for ith sample

as well as

$$\underbrace{\sum_{j=1}^{K} P_{ij} = 1}$$

total prob of ith sample belonging to all classes summation is 1 log likelihood

$$L = P(\vec{Y}|\vec{X};\beta\text{-}k)$$

ith sample $L_i = P(y^{(i)}|x^{(i)};\beta\text{-}k)$

$$= \prod_{j=1}^{k}(P_{ij})^{y_{ij}}$$

$y_i = [0\,0\,0\,1\,0\,0]$
any of col (→ exact 1)
with be a 1
other 0

for all sample

$$L = \prod_{i=1}^{m}\prod_{j=1}^{k}(P_{ij})^{y_{ij}}$$

$$l = \sum_{i=1}^{m}\sum_{j=1}^{k} y_{ij}\log(P_{ij})$$

Parameters are β for which we need to maximize l log 1st ~~last~~ take

$$l_i = \sum_{j=1}^{k} y_{ij}\log(P_{ij})$$

reason

$$\boxed{\sum_{j=1}^{k} y_{ij} = 1}$$

for ith sample

for easy interpretation lets use

$Z_{ij} = X^{(i)}\beta_j$, $P_{ij} = \dfrac{e^{Z_{ij}}}{\sum_{l=1}^{k} e^{Z_{il}}}$

So

$$\frac{\partial l_i}{\partial \beta_t} = \underbrace{\frac{\partial Z_{it}}{\partial \beta_t}}_{x^{(i)}} \times \frac{\partial l_i}{\partial Z_{it}}$$

$$\frac{\partial l_i}{\partial Z_{it}} = \sum_{j=1}^{k}\frac{y_{ij}}{P_{ij}}\frac{\partial P_{ij}}{\partial Z_{it}}$$

$$\frac{\partial P_{ij}}{\partial Z_{it}} = \frac{\delta_{jt}\, e^{Z_{ij}}\sum - e^{Z_{ij}}\times e^{Z_{it}}}{\left(\sum_{l=1}^{k} e^{Z_{il}}\right)^2} \qquad \delta_{jt}\begin{cases}0 & j != t\\ 1 & j == t\end{cases}$$

$$= \frac{(\delta_{jt}\sum - e^{Z_{it}})\,e^{Z_{ij}}}{(\quad)^2} \rightarrow P_{ij} = \frac{(\delta_{jt}\sum - e^{Z_{it}})P_{ij}}{(\sum e^{Z_{il}})}$$

$$\frac{\partial l_i}{\partial Z_{it}} = \frac{\sum_{j=1}^{k} y_{ij}(\delta_{jt}\sum - e^{Z_{it}})}{(\sum e^{Z_{il}})}$$

$$= y_{it} - P_{it} = \sum_{j=1}^{k} y_{ij}\delta_{jt} - P_{it}\sum y_{ij}$$

$$= y_{it} - P_{it}$$

If cost fun^n ⇒ $J(\beta) = -\frac{1}{m}\sum_{i=1}^{m} l_i$

So $\dfrac{\partial l_i}{\partial Z_{it}} = P_{it} - y_{it}$ → scalar

$$\frac{\partial l_i}{\partial \beta_t} = (P_{it} - y_{it})(x^{(i)})^T \in \mathbb{R}^{d\times 1}$$

$$\frac{\partial l}{\partial \beta} = X^T(P-Y)$$

$X \in \mathbb{R}^{m\times d}$
$P-Y \in \mathbb{R}^{m\times k}$

$$\boxed{\frac{\partial J(\beta)}{\partial \beta} = \frac{1}{m}X^T(P-Y)}$$

$X^T \in \mathbb{R}^{d\times m}$
$(P-Y) \in \mathbb{R}^{m\times k}$

$X^T(P-Y) \in \mathbb{R}^{d\times k}$

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$$\frac{\partial J}{\partial \beta} = \frac{1}{m} X^T(P-Y)$$

~~$\frac{\partial J}{\partial \beta_j}$~~ =

BGD => $X^T \in R^{m \times d}$ (1 update) 3 all samples at a time

SGD => $x_i^T \in R^{d \times 1}$ (1 epoch) 1 sample then m updates

Batch => ✓

~~$\frac{\partial J_k}{\partial \beta_j} = \frac{1}{m}\sum_{i=1} (p_{ik} - y_{ik})(x^{(i)})^T$~~

$$\frac{\partial J}{\partial \beta_n} = \sum_{i=1}^{m} (p_{in} - y_{in}) x_i$$